

Dataset:

```
import pandas as pd
```

```
data = {  
    'Student':['A','B','C','D','E','F','G','H'],  
    'Study_Hours':[2,3,4,5,6,7,8,9],  
    'Attendance':[70,75,80,85,90,92,95,98],  
    'Marks':[50,55,60,68,72,78,85,90]  
}
```

```
df = pd.DataFrame(data)
```

2)DATAcleaning:

```
print("Missing Values")
```

```
print(df.isnull().sum())
```

```
df.drop_duplicates(inplace=True)
```

```
print("Data Cleaning Completed")
```

3)data analysis:

```
print(df.describe())
```

#analysis commands:

```
print("Average Marks =", df['Marks'].mean())
```

```
print("Highest Marks =", df['Marks'].max())
```

```
print("Lowest Marks =", df['Marks'].min())
```

```
print("Average Attendance =", df['Attendance'].mean())
```

```
print("Average Marks =", df['Marks'].mean())
```

```
print("Highest Marks =", df['Marks'].max())
```

```
print("Lowest Marks =", df['Marks'].min())
```

```
print("Average Attendance =", df['Attendance'].mean())
```

visualization 2:Marks trend:

```
plt.figure(figsize=(8,5))
```

```
plt.bar(df['Student'],
        df['Attendance'])
```

```
plt.title("Attendance Analysis")
```

```
plt.xlabel("Students")
```

```
plt.ylabel("Attendance (%)")
```

```
plt.show()
```

```
lt.figure(figsize=(6,6))
```

```
plt.pie(
    df['Marks'],
    labels=df['Student'],
    autopct='%1.1f%%'
)
```

```
plt.title("Marks Distribution")
```

```
plt.show()
```

prediction model:

```
from sklearn.linear_model import LinearRegression
```

```
X = df[['Study_Hours']]
```

```
y = df['Marks']
```

```
model = LinearRegression()
```

```
model.fit(X,y)
```

```
prediction = model.predict([[10]])
```

```
print("Predicted Marks for 10 Study Hours =", prediction[0])
```

prediction graph:

```
plt.figure(figsize=(8,5))
```

```
plt.scatter(
    df['Study_Hours'],
    df['Marks']
)
```

```
plt.plot(
```

```
    df['Study_Hours'],
    model.predict(df[['Study_Hours']])
)

plt.title("Study Hours vs Marks Prediction")

plt.xlabel("Study Hours")

plt.ylabel("Marks")

plt.show()
```

Dashboard creation:

```
fig, axs = plt.subplots(2,2, figsize=(12,8))

axs[0,0].plot(df['Student'], df['Marks'])
axs[0,0].set_title('Marks Trend')

axs[0,1].bar(df['Student'], df['Attendance'])
axs[0,1].set_title('Attendance')

axs[1,0].hist(df['Marks'])
axs[1,0].set_title('Marks Distribution')

axs[1,1].scatter(df['Study_Hours'], df['Marks'])
axs[1,1].set_title('Study Hours vs Marks')

plt.tight_layout()

plt.show()
```